

My AI agent runs a language server

— the same brain your IDE uses —

**4 answers grep
can never give you.**

watch it think like a compiler →

1 CATCHES ITS OWN BUGS

It writes code, the language server flags the undefined name, it fixes it, and the error clears. No tests involved.

```
result: get_diagnostics
File: totals.py
[Error] Line 2, Col 16: undefined name 'itms'
```

```
tool: search_replace
{"search":"itms","replace":"items","path":"totals.py"}
```

```
result: search_replace
Applied edit to totals.py:
--- a/totals.py
+++ b/totals.py
@@ -1,2 +1,2 @@
 def total(itms):
-   return sum(itms)
+   return sum(items)
```

```
tool: get_diagnostics
{}
```

```
result: get_diagnostics
No diagnostics reported.
```

2 FOLLOWS THE IMPORT

api.py calls save_user. It answers with the file and line where it's really defined — db.py, line 4.

```
tool: find_definition  
{"path":"api.py","line":5,"character":11}
```

```
result: find_definition  
File: db.py  
Start: Line 4, Col 5  
End: Line 4, Col 14
```

```
tool: finish  
{"summary":"save_user is defined in db.py:4."}
```

```
result: finish  
save_user is defined in db.py:4.
```

3 FOLLOWS AN ALIAS

handlers.py calls persist. The word “persist” appears zero times in db.py — the server still lands on line 4.

```
tool: find_definition  
{"path":"handlers.py","line":4,"character":11}
```

```
result: find_definition  
File: db.py  
Start: Line 4, Col 5  
End: Line 4, Col 14
```

```
tool: finish  
{"summary":"persist is defined in db.py at line 4, character 5."}
```

```
result: finish  
persist is defined in db.py at line 4, character 5.
```

4 FINDS EVERY USE

5 references across 3 files — including one hiding in a type annotation. Grep would miss the shape of it.

```
tool: find_references  
{"character":6,"path":"models.py","line":0}
```

```
result: find_references  
File: api.py, Line 1, Col 20  
File: api.py, Line 6, Col 22  
File: db.py, Line 1, Col 20  
File: db.py, Line 4, Col 18  
File: models.py, Line 1, Col 7
```

Part 5 — earlier: “8 things a local AI coding agent can do” · “I gave an AI agent shell access” · “What makes an AI coding agent production-grade?” · “Your AI agent never reads your code”

Want to build one?

This entire agent is a book —
and a free, open-source repo.



Get the book (Kindle)

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Clone the code (MIT)

github.com/Natarajan-R/local-ai-coding-agent

Follow for the build, one file at a time.

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